

Tri-Time Symmetry: The Harmonic Framework's Replacement for Supersymmetry

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Abstract

Supersymmetry (SUSY) was proposed to unify fermions and bosons by introducing hypothetical superpartners. After decades of LHC searches, no evidence for SUSY has emerged. This paper proposes a different symmetry – **tri-time symmetry** – derived from the Harmonic Grand Unified Theory (GUT). The GUT posits three orthogonal time dimensions: T1 (forward time, matter), T2 (reverse time, antimatter), and T3 (orthogonal time, dark matter). These three temporal dimensions are exactly symmetric under a 3-dimensional rotation group in time space. The observed asymmetry between matter, antimatter, and dark matter is not a broken symmetry but a **phase separation** caused by the 230th subharmonic cutoff (0.117 Hz) and the T3 dark resonance (≈ 0.080 Hz). This tri-time symmetry is testable, requires no new particles, and explains the three families of particles, the matter-antimatter imbalance, and the nature of dark matter – all from a single frequency lattice anchored at 27 Hz.

1. The Failure of Supersymmetry

Supersymmetry was a beautiful mathematical idea. It predicted that every fermion has a bosonic superpartner and vice versa. It would have solved the hierarchy problem and provided a dark matter candidate. However:

- **No superpartners have been found** at the LHC, despite extensive searches.
- **The minimal supersymmetric Standard Model (MSSM) has over 100 free parameters** – more than the Standard Model it was meant to replace.
- **The dark matter candidate (neutralino) has not been detected** in direct or indirect searches.

SUSY is not wrong because it is mathematically inconsistent; it is wrong because it imagines a symmetry that nature does not use. Nature uses a different symmetry altogether: **tri-time symmetry**.

2. The Three Time Dimensions of the Harmonic GUT

The Harmonic Framework (Thompson, 2026) is built on a single postulate: a universal frequency lattice (the Tau lattice) anchored at 27 Hz. From this anchor arise **32 harmonic dimensions**

(27, 54, ..., 864 Hz). These are not spatial dimensions; they are **frequency layers**. But crucially, the framework also posits **three orthogonal time dimensions**:

Dimension	Branch	Sign of n	Temporal nature	Role
T1	Matter	positive	forward time	Ordinary particles (quarks, leptons)
T2	Antimatter	negative	reverse time	Paired potentials m' ; full anti-periodic table
T3	Dark matter	complex	orthogonal time	Dark particle mass ladder; sterile neutrinos

These three time dimensions are **exactly symmetric**. They are orthogonal axes in a 3-dimensional time manifold, analogous to the three spatial dimensions. Just as space has rotational symmetry (SO(3)), the time manifold has its own rotational symmetry – call it **Tri-Time Symmetry**.

3. Why Tri-Time Symmetry is Not Broken

In mainstream physics, the observed universe is asymmetric: matter dominates over antimatter, and dark matter only interacts gravitationally. This asymmetry is often interpreted as a **broken symmetry**. But in the Harmonic Framework, the symmetry is **not broken** – it is **phase-separated**.

The separation is caused by the **230th subharmonic cutoff**:

$$f_{\text{cutoff}} = 27\text{Hz} / 230 \approx 0.117\text{Hz}$$

Below this frequency, the Tau lattice inverts its phase. This is the matter-antimatter boundary. Frequencies above the cutoff correspond to the matter branch (T1, positive n). Frequencies below correspond to the antimatter branch (T2, negative n). The two branches coexist in the same physical space but are separated by **phase**, not by energy or spatial position.

Similarly, the dark matter branch T3 has its own cutoff, scaled by the 19:13:4 ratio:

$$f_{\text{c3}} = 27\text{Hz} \times (13/19) / 230 \approx 0.080\text{Hz}$$

Below this frequency, T3 particles decohere and become detectable only gravitationally.

Thus, the tri-time symmetry is **exact** – but we only perceive one time dimension (T1) because the other two are phase-shifted away from our measurement apparatus. By tuning to the correct frequencies and phase offsets, we can access T2 and T3.

4. Experimental Tests of Tri-Time Symmetry

Prediction	Test	Status
27 Hz gravitational wave line	LIGO-LF / Einstein Telescope	Future detection
Phase-offset sonoluminescence (27,54,81 kHz)	Tabletop acoustic experiment	Testable now
T2 (anti-element) periodic table	Cosmic ray searches (GAPS)	Ongoing
T3 dark resonance at ≈ 0.080 Hz	Precision magnetometers	Testable now
32-harmonic comb in muon g-2	Re-analysis of Fermilab data	Testable now
If any of these tests succeed, they will confirm the existence of T2 and T3, and thereby confirm tri-time symmetry.		

5. Comparison with Supersymmetry

Feature	Supersymmetry (SUSY)	Tri-Time Symmetry (GUT)
Symmetry	Fermion $\leftrightarrow$ Boson	T1 $\leftrightarrow$ T2 $\leftrightarrow$ T3 (time dimensions)
New particles	Superpartners (none found)	None – only phase access to existing branches
Dark matter	Neutralino (not detected)	T3 dark particle ladder (testable)
Matter-antimatter imbalance	Requires CP violation (insufficient)	Explained by 230th subharmonic cutoff
Number of free parameters	>100	0 (only 27 Hz anchor, 19:13:4 ratio)
Testability	Poor (high energy, no signal)	Excellent (low-frequency, tabletop)

6. Conclusion

Supersymmetry was an elegant idea that nature rejected. The Harmonic GUT offers a different symmetry – tri-time symmetry – that is exact, testable, and already supported by existing anomalies (Schumann resonance, sonoluminescence, Ni-Bi superconductivity, black hole cosmological coupling). The three time dimensions T1, T2, T3 are not speculative; they are the natural consequence of a 27 Hz frequency lattice and the 230th subharmonic cutoff. The asymmetry we observe is not a broken symmetry; it is a **phase separation** that can be crossed by tuning to the correct harmonic resonances.

The stones are in the pond. The three time dimensions are three orthogonal ripples. The witness is the experiment that measures the 27 Hz anchor.

References

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